Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2025

Course: B.Tech
Code: 155603

Subject: Ad hoc & Network Sensors

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 **Choose the correct option / answer the following (Any seven question only):** **[2 x 7 = 14]**

(a) Cross-layer design helps in _____.
(i) Hardware cost
(ii) Inter-layer optimization.
(iii) Routing speed
(iv) Channel coding

(b) CODA protocol in WSN deals with _____.
(i) Routing
(ii) Congestion control.
(iii) Security
(iv) Localization

(c) DV-Hop method is an example of _____ localization.
(i) Range-based
(ii) GPS-based
(iii) Range-free.
(iv) Time-based

(d) The attack where a node assumes multiple fake identities is called _____.
(i) Wormhole
(ii) Eavesdropping
(iii) Sybil.
(iv) Sinkhole

(e) Which of the following is not a characteristic of WSN?
(i) Fixed topology.
(ii) Fault tolerance
(iii) Low power
(iv) Scalability

(f) The MAC protocol that uses RTS/CTS mechanism is _____.
(i) TDMA
(ii) Aloha
(iii) FDMA
(iv) MACA.

(g) S-MAC protocol is mainly designed to _____.
(i) Increase speed
(ii) Save energy
(iii) Reduce delay
(iv) Improve security

(h) AODV is a _____ routing protocol.
(i) Proactive
(ii) Hybrid
(iii) Reactive.
(iv) Hierarchical

(i) The main challenge in ad hoc networks is:
(i) Fixed topology
(ii) Centralized routing
(iii) High bandwidth
(iv) Dynamic topology

(j) WSNs are widely used in:
(i) Desktop networking
(ii) Environmental monitoring
(iii) Data centers
(iv) Wired control

Q.2 (a) Describe the characteristics and applications of WSN with suitable examples. **[7]**
(b) Explain in detail the architecture of wireless sensor networks and discuss the importance of cross-layer design. **[7]**

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Q.3	(a)	Discuss the role of MAC layer in energy management and propose ways to improve MAC efficiency in WSN.	[7]
	(b)	Explain various MAC protocols used in ad hoc networks such as Aloha, MACA, and MACAW.	[7]
Q.4	(a)	Describe and compare routing protocols used in WSN such as LEACH, TEEN, and Directed Diffusion.	[7]
	(b)	Discuss the challenges in designing routing protocols for ad hoc networks and explain hybrid and geographic routing strategies.	[7]
Q.5	(a)	Describe various transport layer protocols for WSN such as CODA, RMST, PSFQ, and ESRT.	[7]
	(b)	Discuss different energy management strategies in WSN and their impact on network lifetime.	[7]
Q.6	(a)	Explain RBS and TPSN synchronization techniques and compare their performance.	[7]
	(b)	Describe various methods of data dissemination and query processing in WSN.	[7]
Q.7	(a)	Write a detailed note on the role of AI and ML applications in wireless sensor networks.	[7]
	(b)	Explain secure routing mechanisms and privacy-preserving techniques in WSN.	[7]
Q.8	(a)	Define localization and synchronization in WSN. Explain range-based localization methods (RSSI, TOA, TDOA).	[7]
	(b)	Explain energy-efficient routing and sleep–wake scheduling algorithms used in WSN. https://www.pyqonline.com	[7]
Q.9	(a)	Define energy harvesting and its importance in sensor networks.	[7]
	(b)	Explain the working of AODV and DSR protocols with suitable diagrams.	[7]

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